GIS Data Model

- The geographical variations needs to be converted into discreet or continuous features to facilitate storage and processing

- Data Models are rules and methods which define how spatial features are represented in a GIS

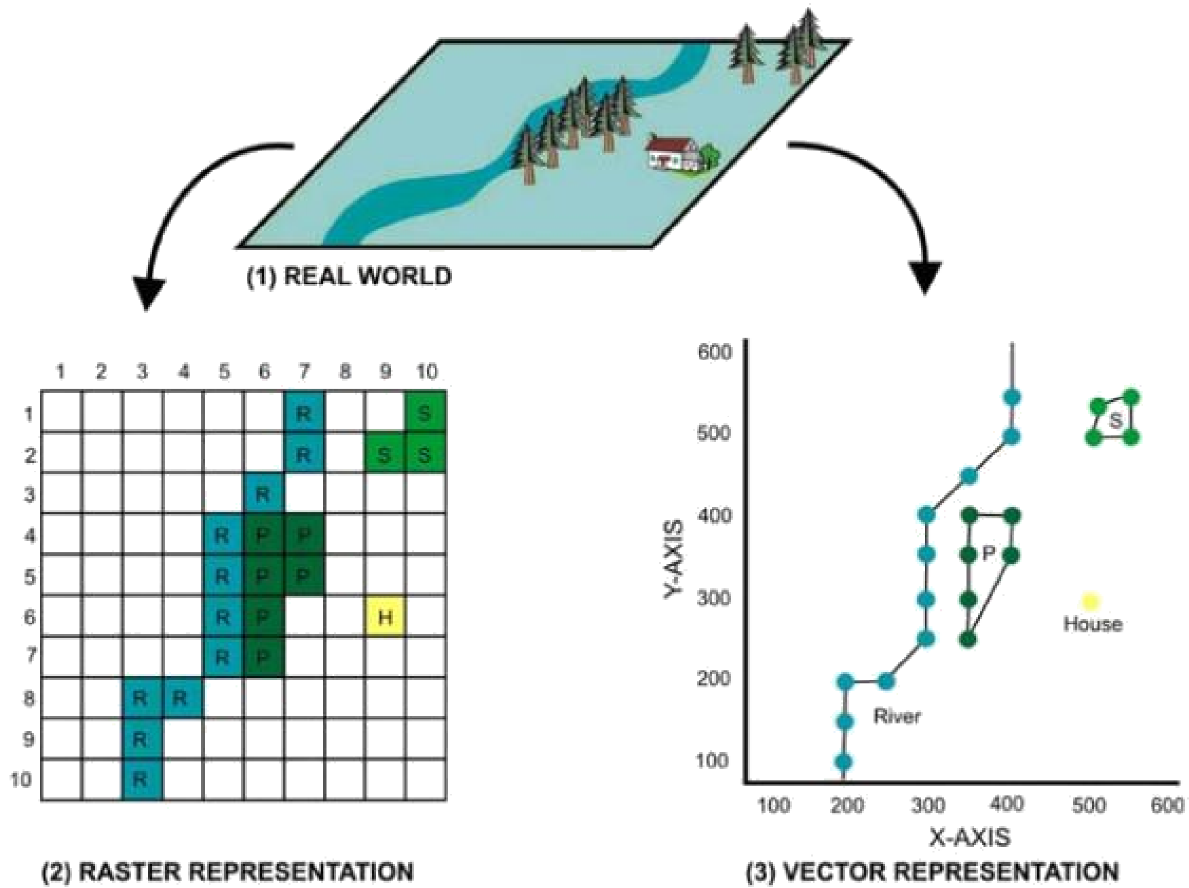
- Data model can be defined as “a set of guidelines for the representation of the logical organization of the data in a database… (consisting) of named logical units of data and the relationships between them”

  - Tsichritzis and Lochovsky (1977)

The two data models used in GIS are:

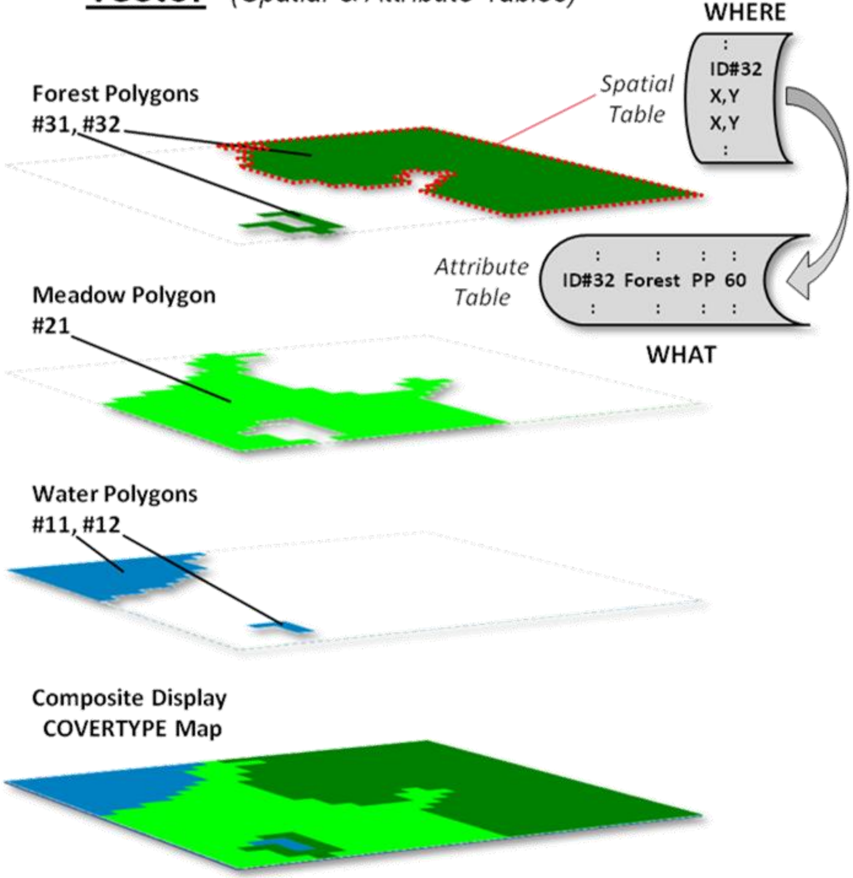
- **Raster**
- **Vector**

Raster and Vector Data Model - Overview

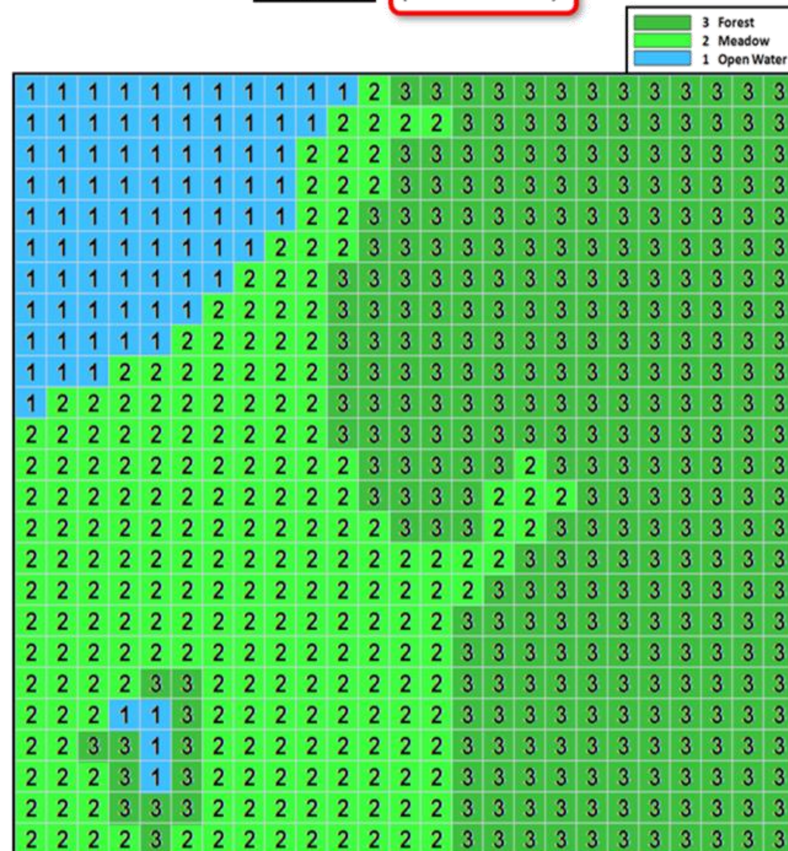


Raster and Vector Data Model - Overview

Vector *(Spatial & Attribute Tables)*



Raster *(Grid Matrix)*



Mapping/Geo-Query *(Discrete Spatial Objects)*

Where – a Spatial Table contains X,Y coordinates delineating the location of each point, line and polygon boundary

What – a linked Attribute Table contains text/values indicating the classification of each spatial object

(Continuous Map Surfaces) **Map Analysis/Modeling**

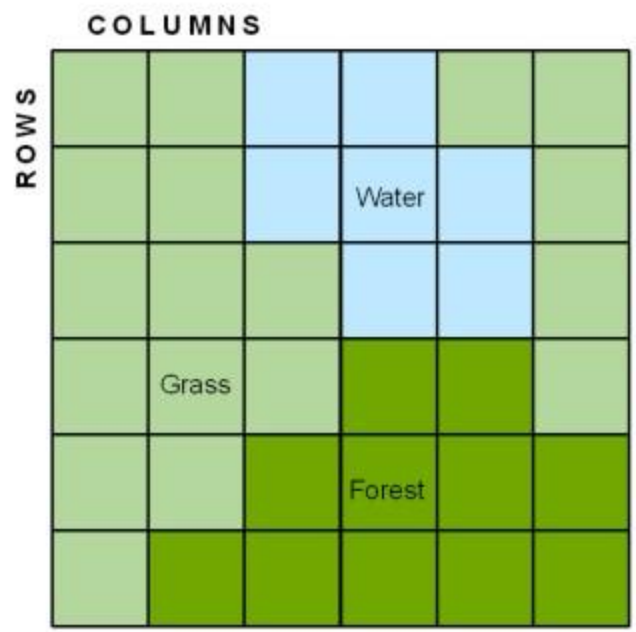
Where – Cell Position in the matrix determines location within a continuous, regular grid

What – Cell Value in the matrix indicates the classification at that location

Raster Data Model

- Raster Model is a representation of the study area (area of interest) as a surface divided into a regular grid of cells (or pixels) and each pixel has an associated value

- To represent the raster visually, each pixel value is assigned to a particular colour



Elements of Raster Data Model

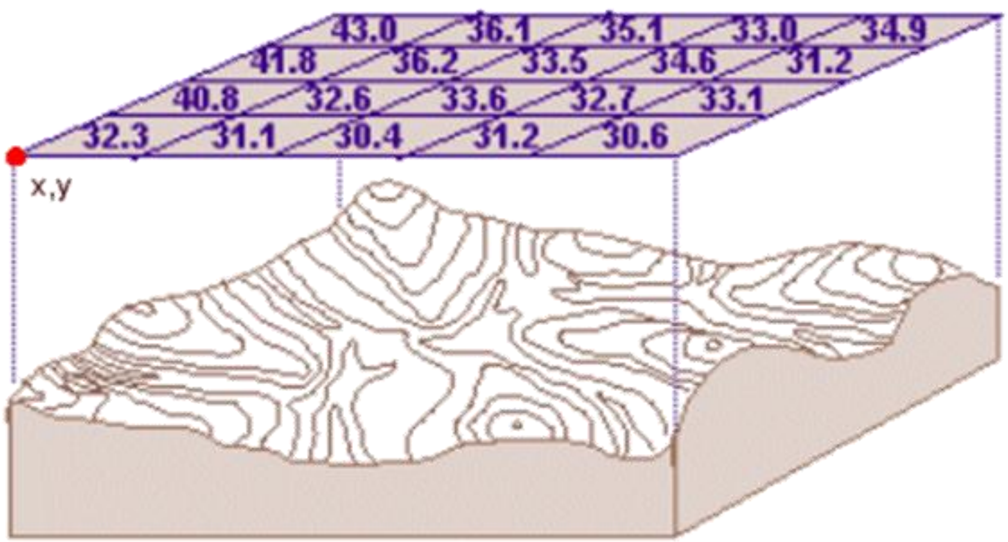
Cell Value

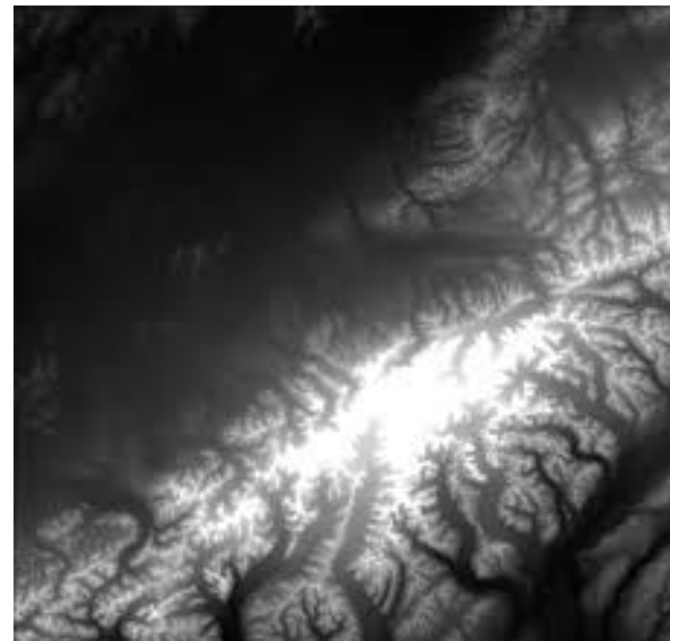
- Each cell in a raster contains a value (called pixel value or DN value), which represents the characteristics of any spatial phenomena occurring at the location represented by the cell (or pixel)

- Can be:
    - Categorical Values *(like 0 for water and 1 for land)*
    - Between 0 - 255 (8-bit value) *for Remote Sensing Image*
    - Integers
    - Floating point values
    - Real Values *(like real elevation values in DTM)*

- A pixel cannot have more than one value

Cell Value

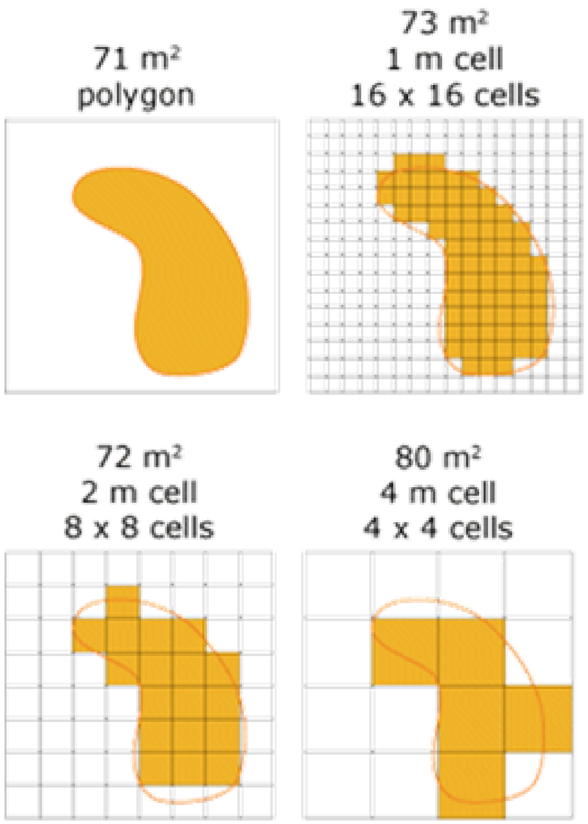


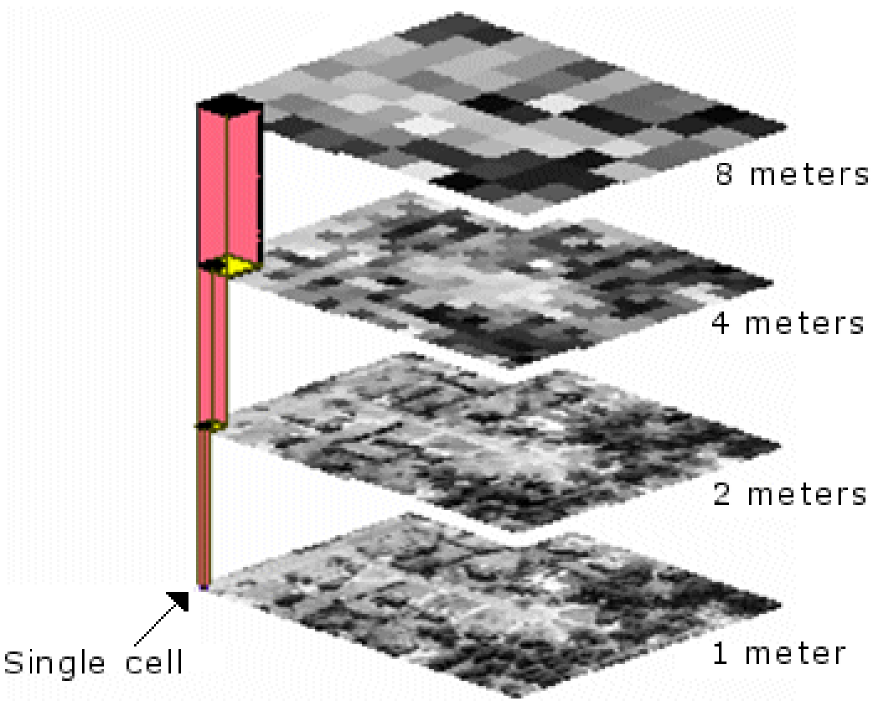


Eg: Digital Elevation Model (DEM)
gives the elevation of the earth's surface

Cell Size (Spatial Resolution)

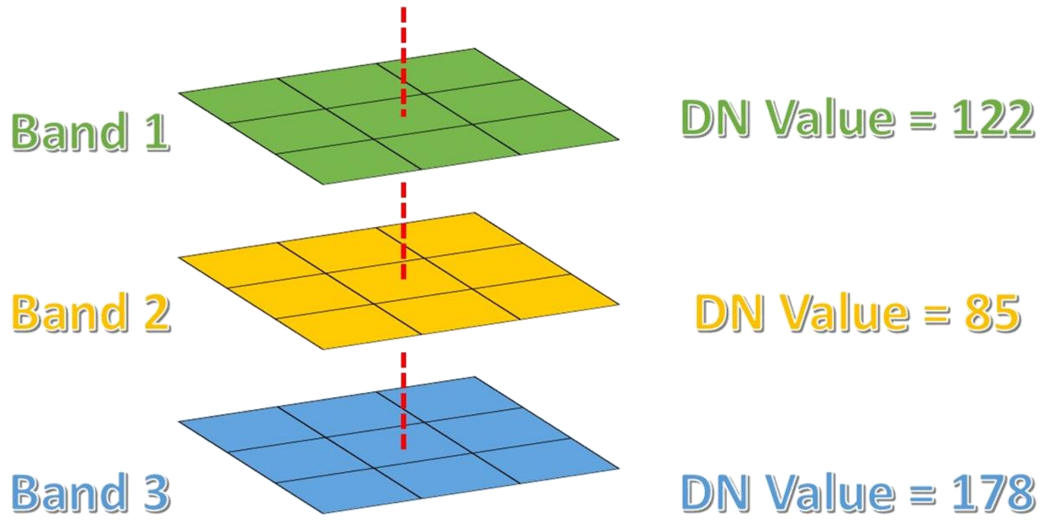
- Cell size determines the resolution of the raster data model
- Spatial resolution refers to the dimension of the cell size representing the area covered on the ground



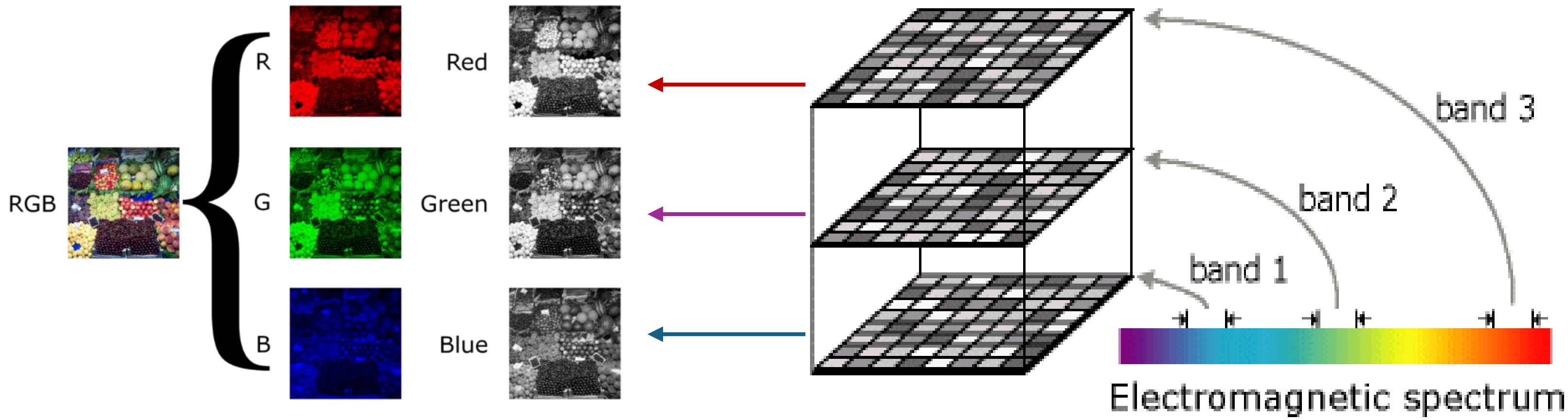


Raster Bands

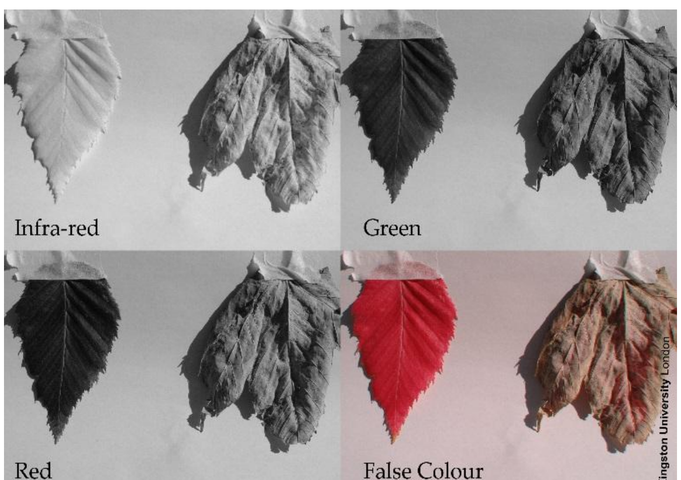
- To represent multiple values per cell (or pixel), the raster needs to have multiple bands



True Colour Images



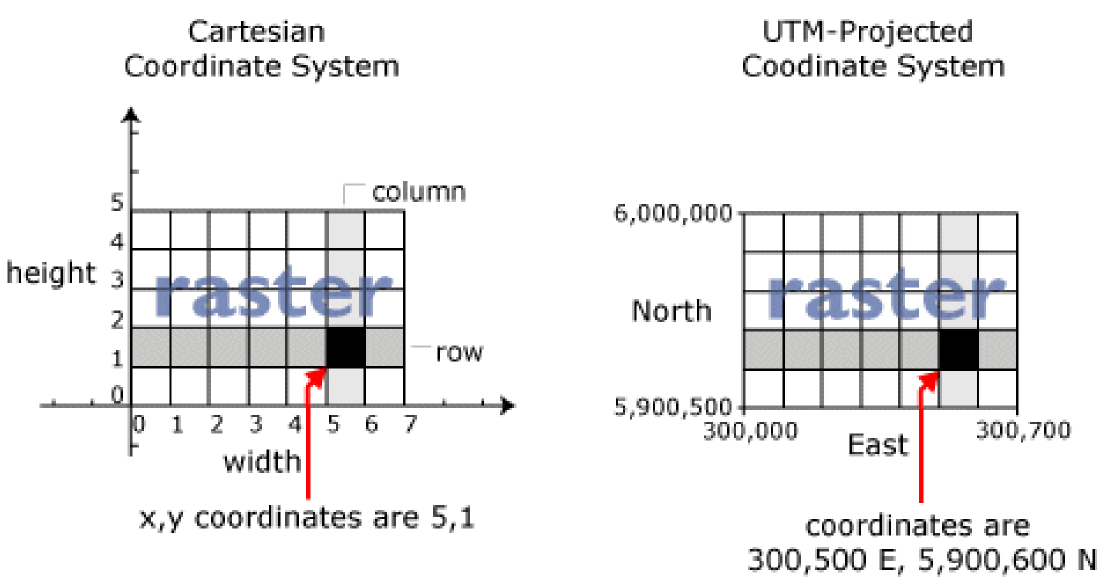
False Colour Images

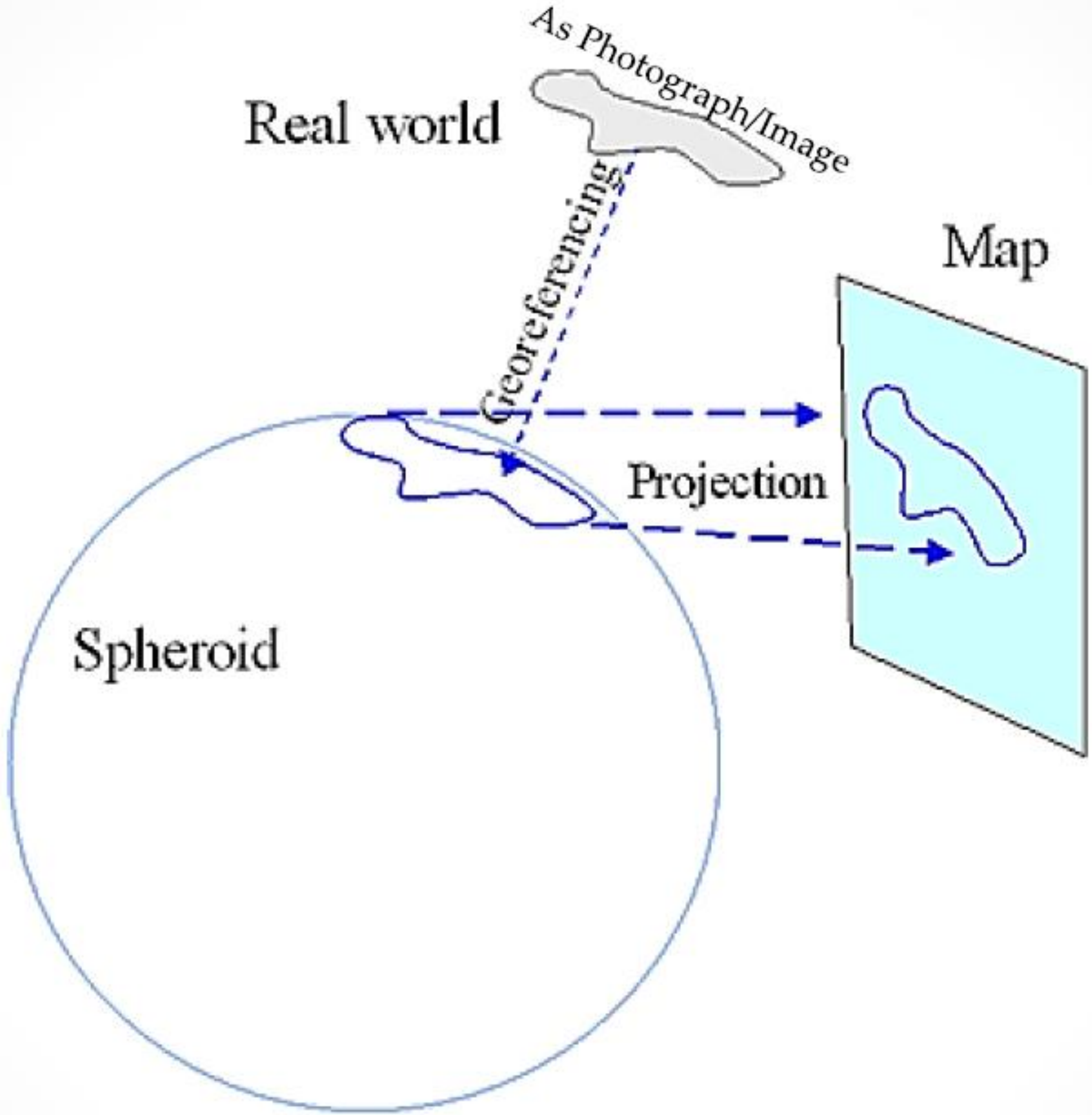


Spatial Reference

- Any cell of a raster can be identified by an ordered pair of coordinates (row and column numbers)

- Generally a projected coordinate system is associated with the raster's own coordinate system which locate each raster cell in the geographic space. Such raster which has been processed to match a projected coordinate system is often called a georeferenced raster




As Photograph/Image
Real world
Georeferencing
Map
Projection
Spheroid

Raster File Organization

Storage of Raster Data (*without any compression*):

- Cell values are stored in a file stored in a file as rows and columns in the form of a matrix

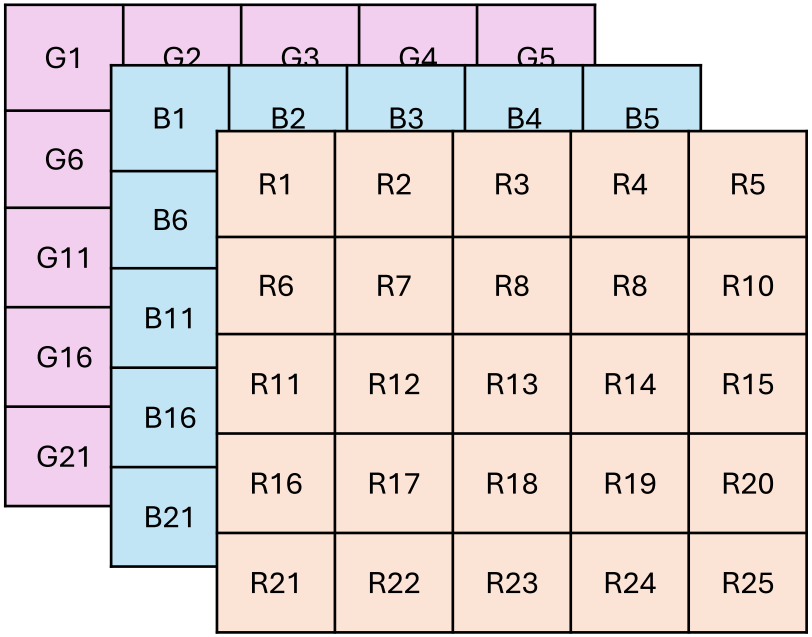
A1	A2	A3	A4	A5
A6	A7	A8	A8	A10
A11	A12	A13	A14	A15
A16	A17	A18	A19	A20
A21	A22	A23	A24	A25

Raster

A1	A2	A3	A4	A5
A6	A7	A8	A8	A10
A11	A12	A13	A14	A15
A16	A17	A18	A19	A20
A21	A22	A23	A24	A25

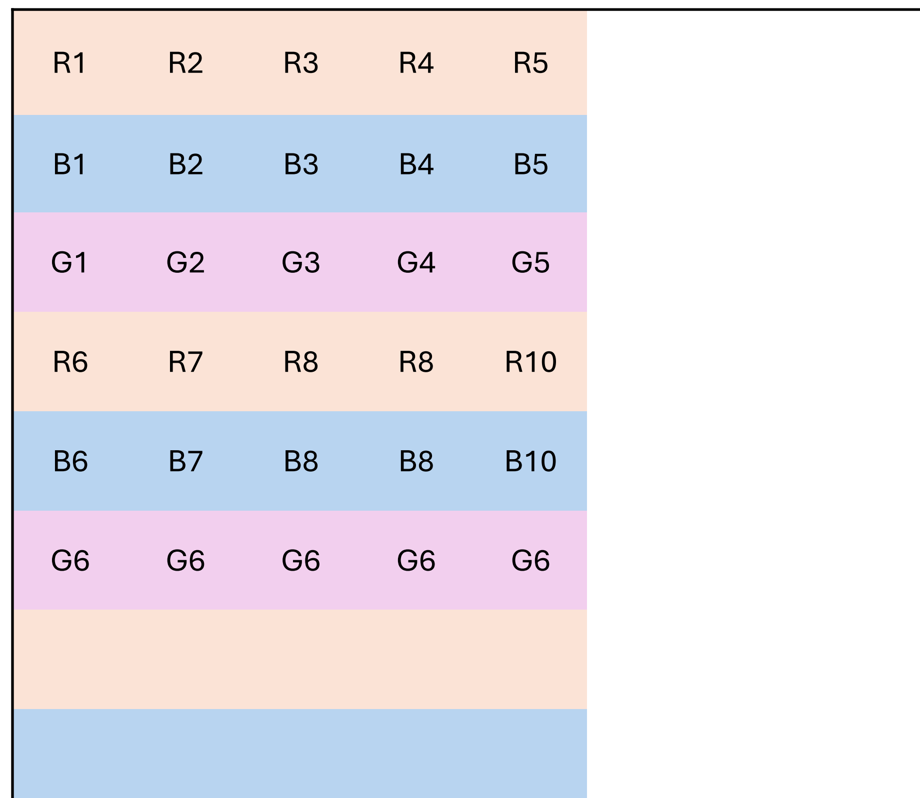
Cell values organization in a File

Band Interleave by Line (BIL)



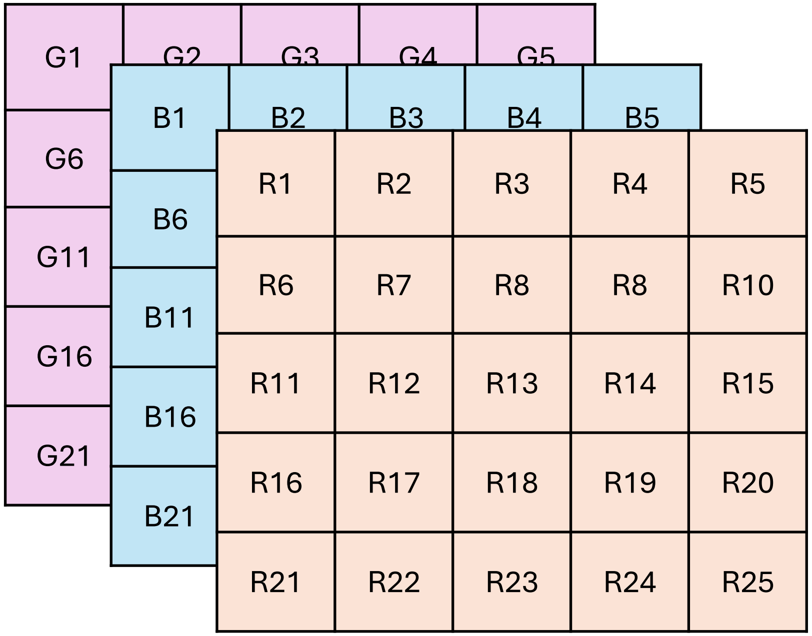
3 Band Raster

Band interleaved by line data stores pixel information band by band for each line, or row



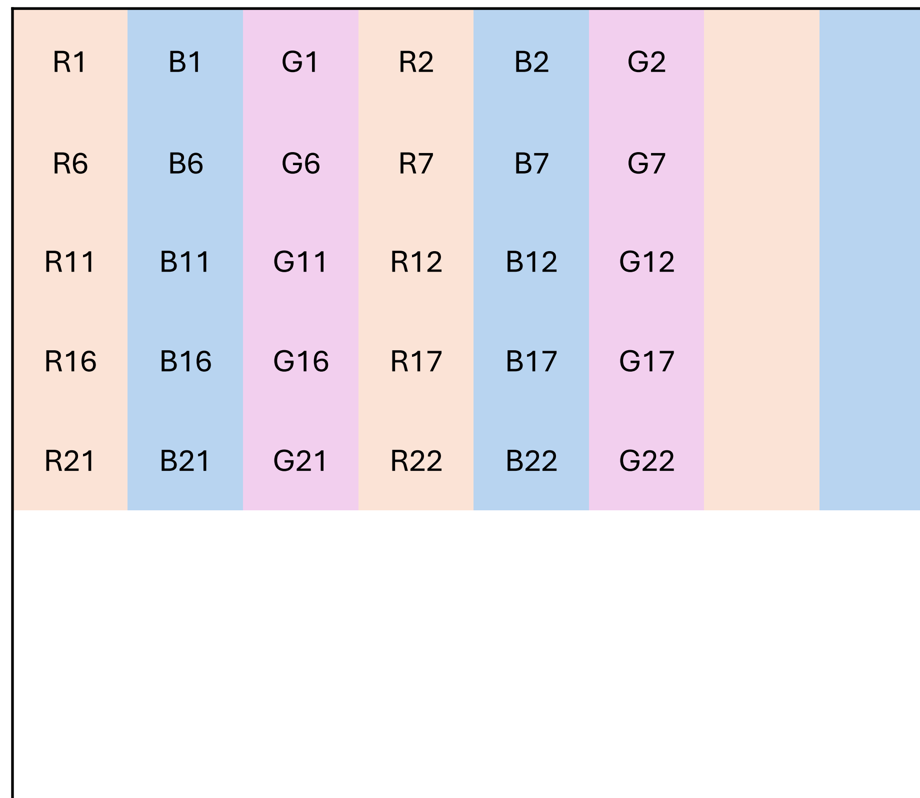
Cell values organization in a File

Band Interleave by Pixel (B I P)



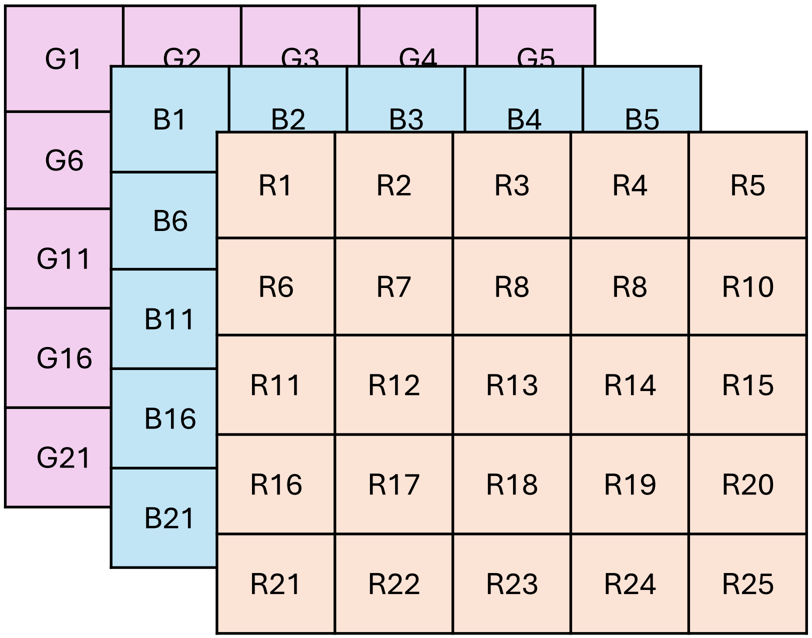
3 Band Raster

In band interleaved by pixel, each pixel is written band by band



Cell values organization in a File

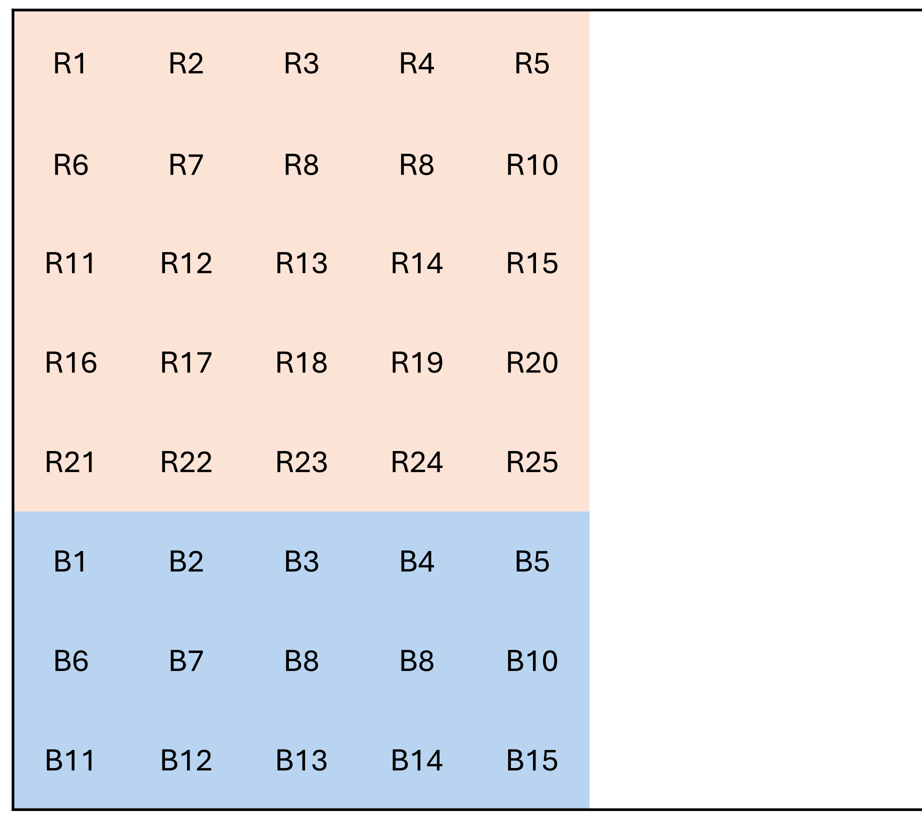
Band Sequential (BSQ)



3 Band Raster

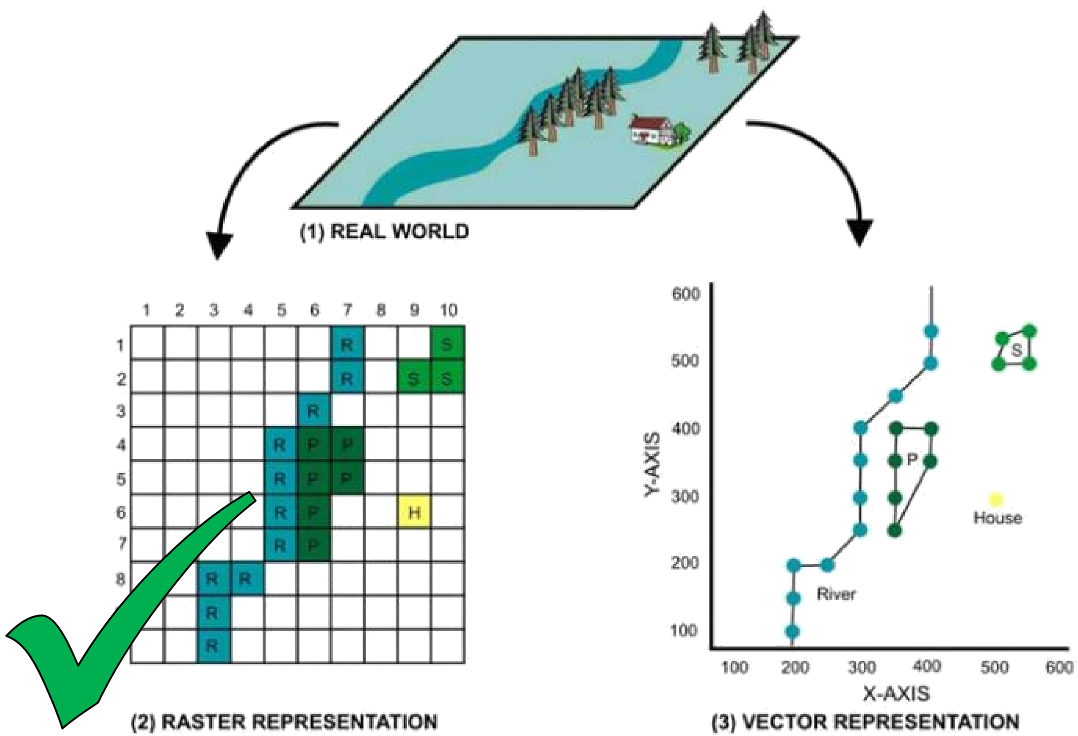
Band sequential format stores information for one band at a time

Cell values organization in a File



Quick RECAP

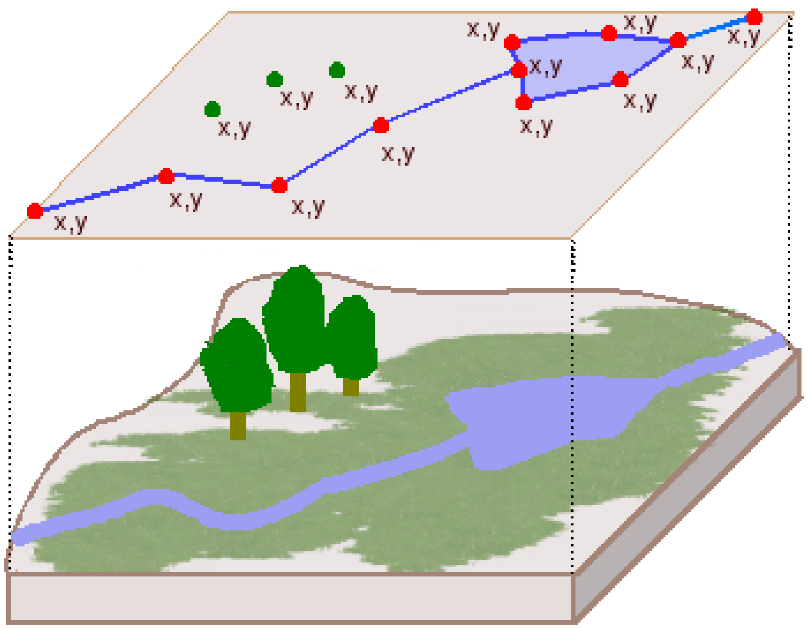
- **GIS Data Model**



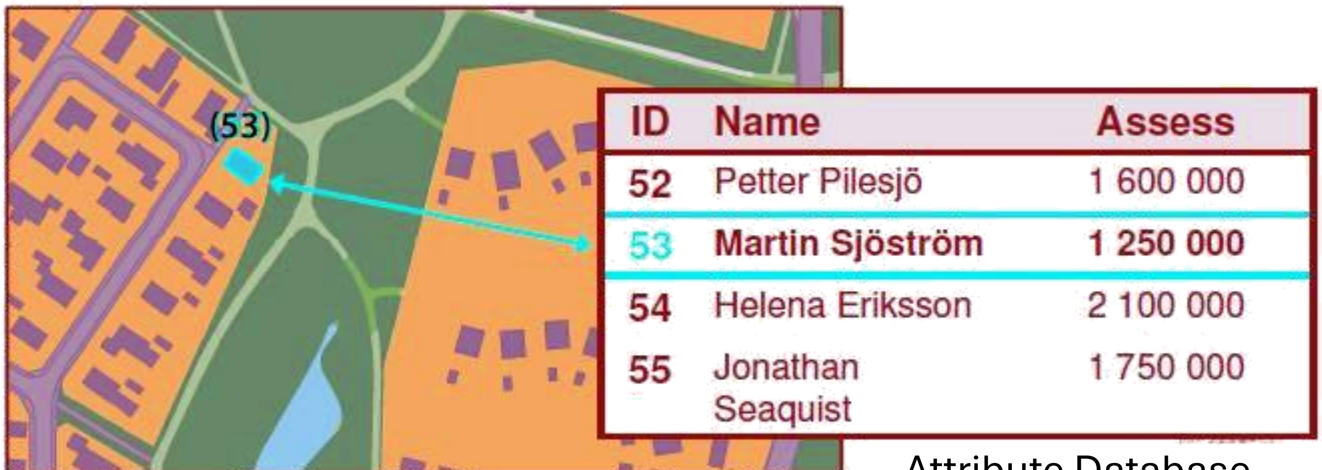
Vector Data Model

- Vector Model represents the area of interest as a combination of primitive spatial data objects

- These primitive spatial data objects are ***Points***, ***Lines*** and ***Polygons*** which are located by a pair of x, y coordinates in a spatial reference frame



- Traditional vector data model stores the spatial and attribute information separately in a ***graphics file*** and ***relational database*** respectively



ID	Name	Assess
52	Petter Pilesjö	1 600 000
53	**Martin Sjöström**	**1 250 000**
54	Helena Eriksson	2 100 000
55	Jonathan Seaquist	1 750 000

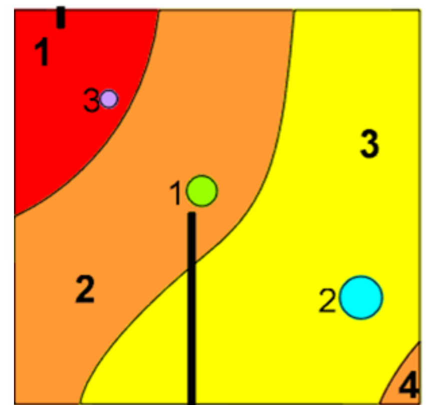
Graphics File

Attribute Database

- The spatial information and attribute information for these models are linked via a simple identification number that is given to each feature in a map.

Points

- 0 – dimension
- Has the property of Location
- Represented by one or more pair of (x, y) coordinates
- Other point features: *node* or *vertex*



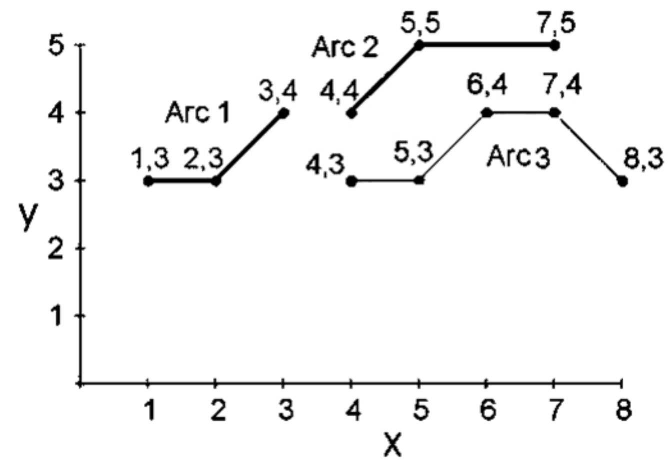
Point ID	X-coord.	Y-coord.
1	3	3
2	4	2
3	2	5

Statione	Sedimentation (cm ka^{-1})	Point ID
MT4	1.5	1
MT7	2.4	2
SS22	0.3	3

Attribute Table

<u>Lines</u>

- 1 – dimension
- Has the property of Length
- Two end-points are required to represent a simple line feature. There can be more points in between these two
- Can be called *edge*, arc, *link* or *chain*



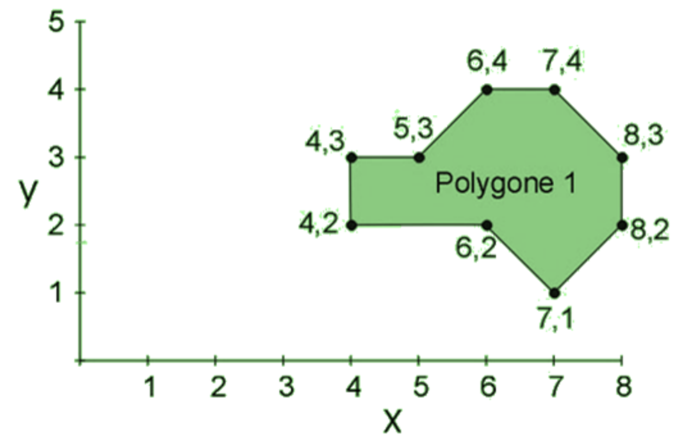
Arc ID - Nr	x,y coordinates
1	1,3 2,3 3,4
2	4,4 5,5 7,5
3	4,3 5,3 6,4 7,4 8,3

Arc ID	Name	Type
1	NH 5	3 Lane
2	NH 2	6 Lane

Attribute Table

Polygon

- 2 – dimension
- Has the property of Area and Perimeter
- Can be defined with a set of coordinates or arc/lines which represents a complete loop
- Can be called *area* or *zone*

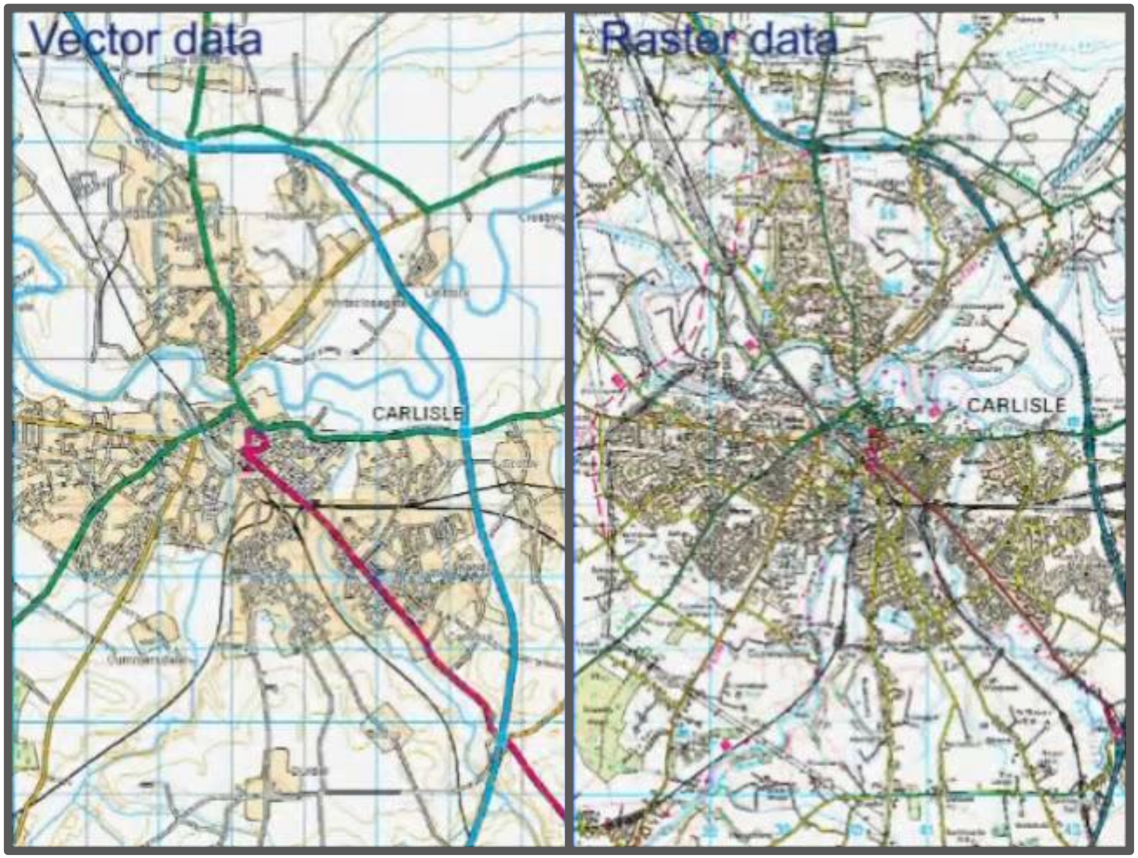


Polygone ID - Nr	x,y coordinates
1	4,2 7,1 8,2 8,3 7,4 6,4 5,3 4,3 4,2

Poly ID	Plot No.	Owner
1	B-34	xyz
2	B-76	abc

Attribute Table

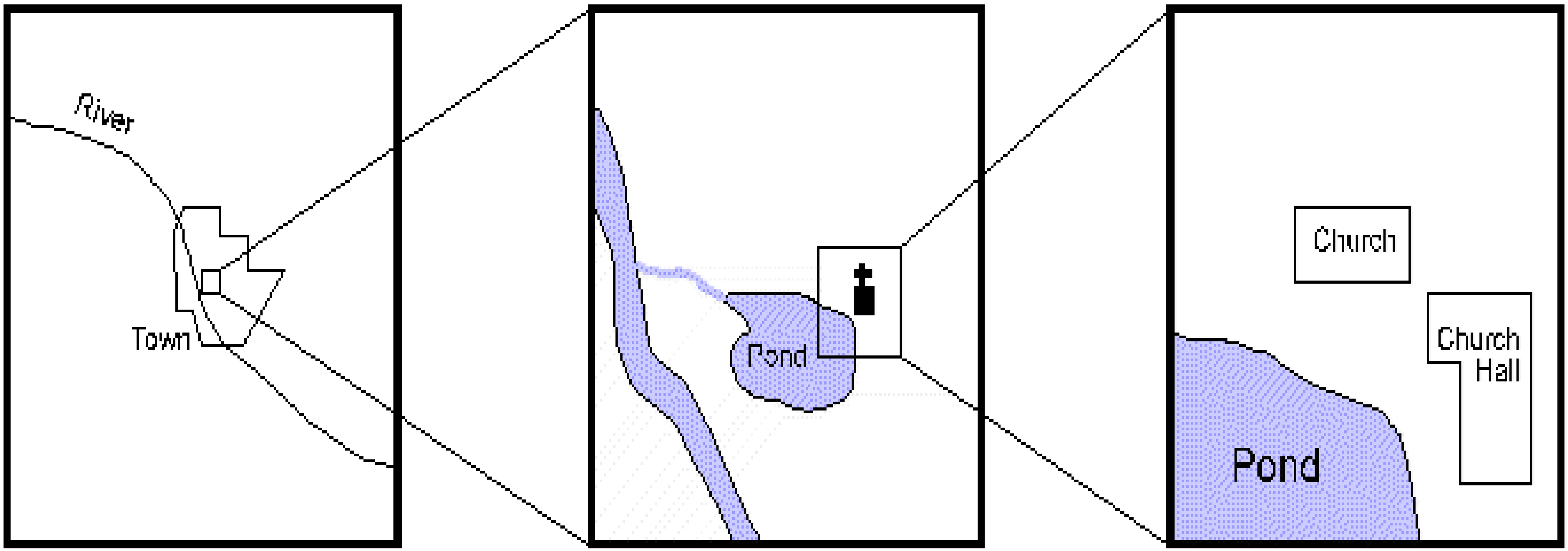
Zooming on Raster and Vector data



Vector Data　　　　　　　　Raster Data

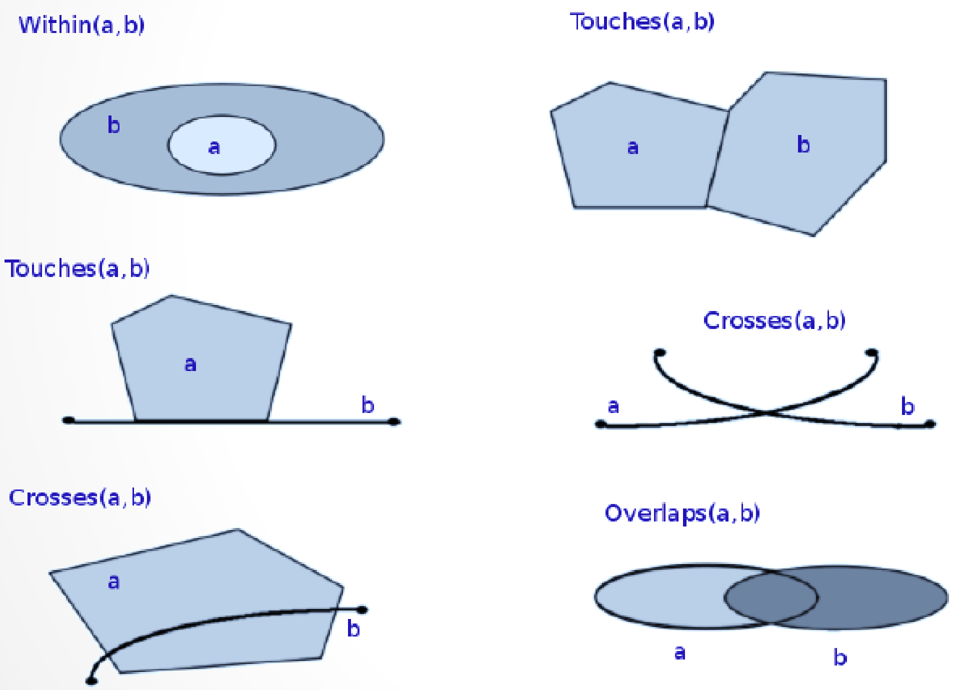
Representation Problem

- Same geographical phenomenon can have different representations at different scales and with different levels of accuracy



Spatial Relation

- A spatial relation, specifies how some object is located in space in relation to some reference object.

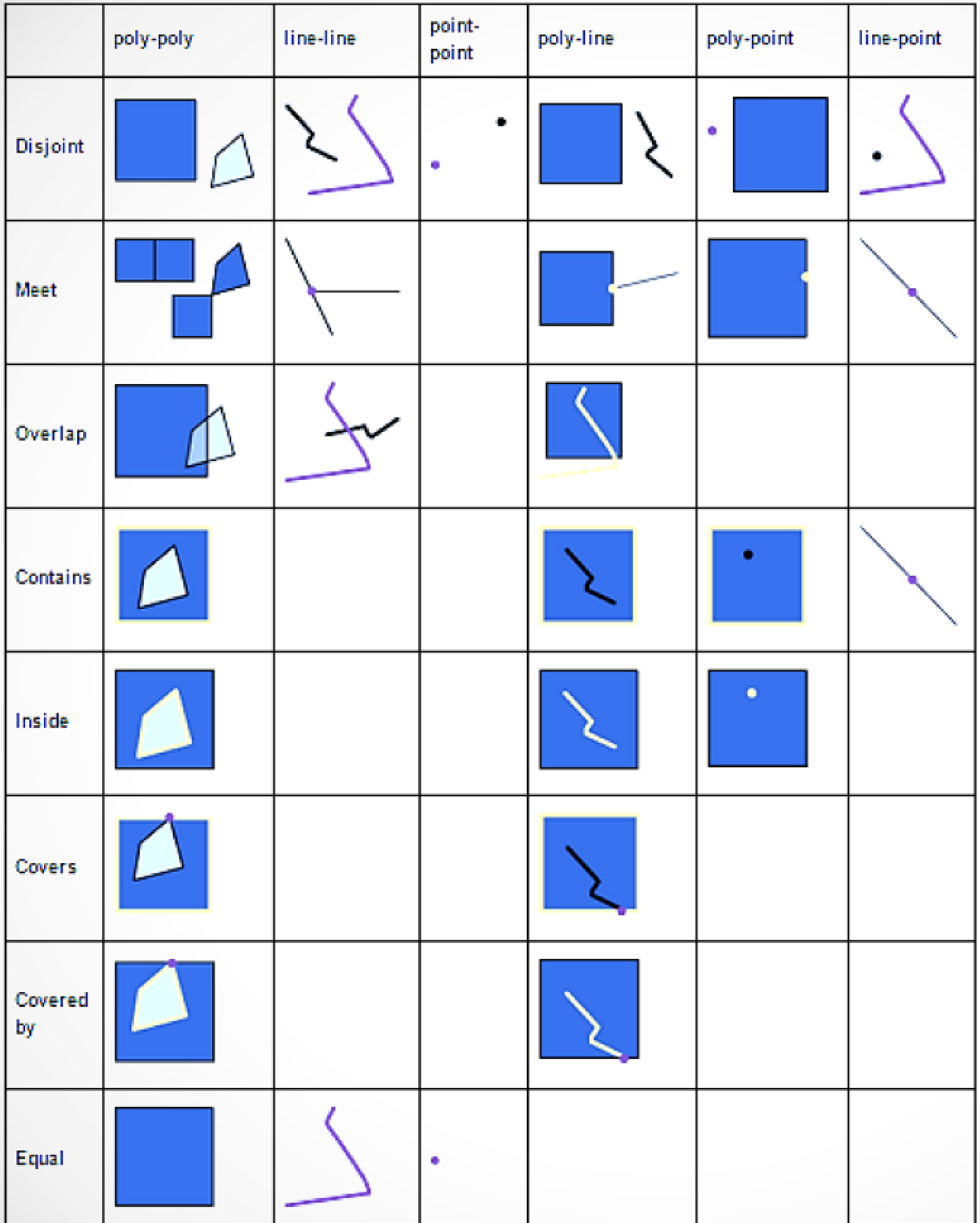


+

Directional Relations

+

Distance Relations



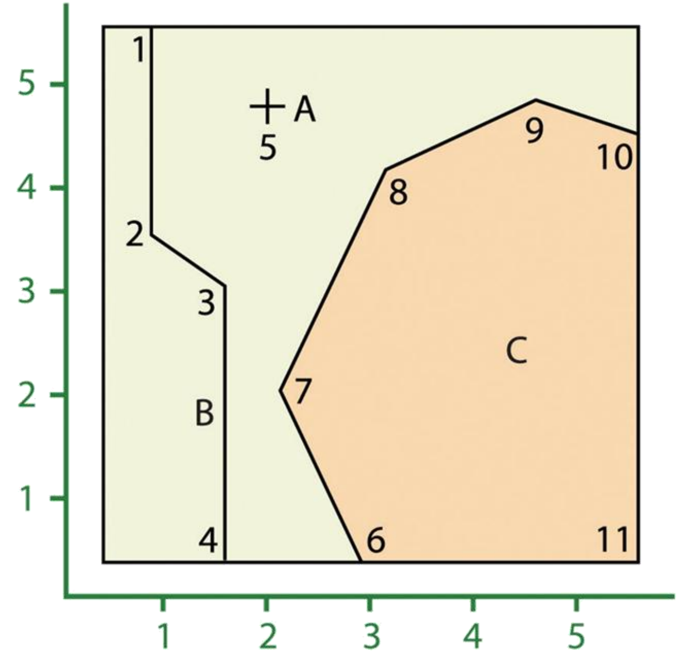
Examples of spatial relationship between features

Vector

- Instead of trying to keep track of the millions of tiny pixels in a raster image, vector images keep track of points and the equations for the lines that connect them. Generally speaking, vector images are made up of paths or line art that can infinitely scalable because they work based on algorithms rather than pixels.
- One of the greatest things about vector images is that you can re-size them infinitely larger or smaller, and they will still print out just as clearly, with no increase (or decrease) in file size. If you remember back to your high school geometry, the equation for a circle of center (h,k) and radius r is $(x - h)2 + (y - k)2 = r2$. If you want to make the circle bigger, you just increase the value of r - instead of having to keep track of tons more pixels, the computer just has to keep track of a different number. That takes almost no file space at all.

Vector Data Structures

- **Spaghetti data model**
  - Simplest vector data structure
  - Each point, line, and/or polygon feature is represented as a string of X, Y coordinate pairs



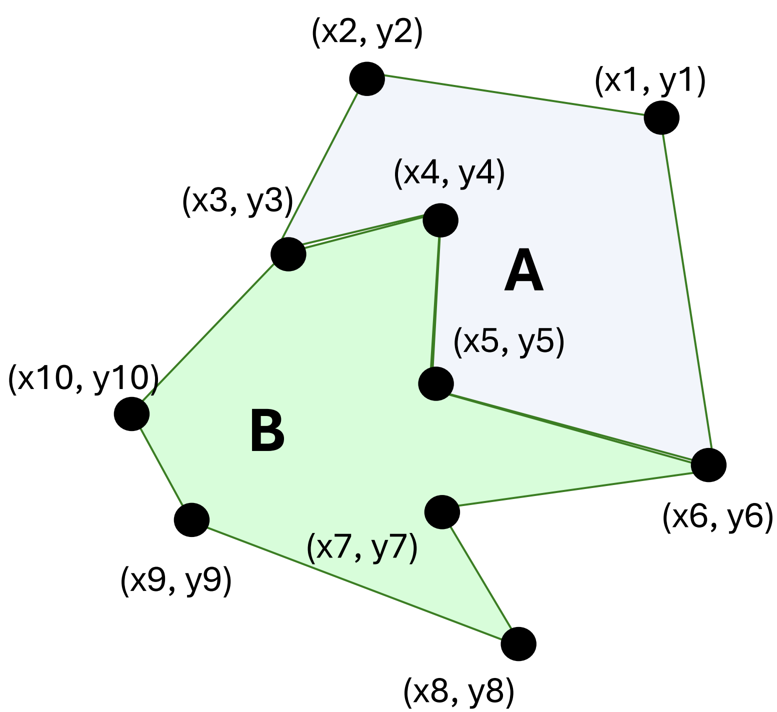
A,1 (identifier of the point and the number of vertex)
2, 4.7

B, 4 (identifier of the line and number of vertex)
1, 5.6
1, 3.5
1.6, 3
1.6, 0.4

C, 6 (identifier of polygon and number of vertex)
3, 0.4 (coordinates of the first vertex)
2.2, 2
3, 4.3
4.8, 4.8
5.6, 4.4
5.6, 0.4
3, 0.4 (coordinates of the first vertex again)

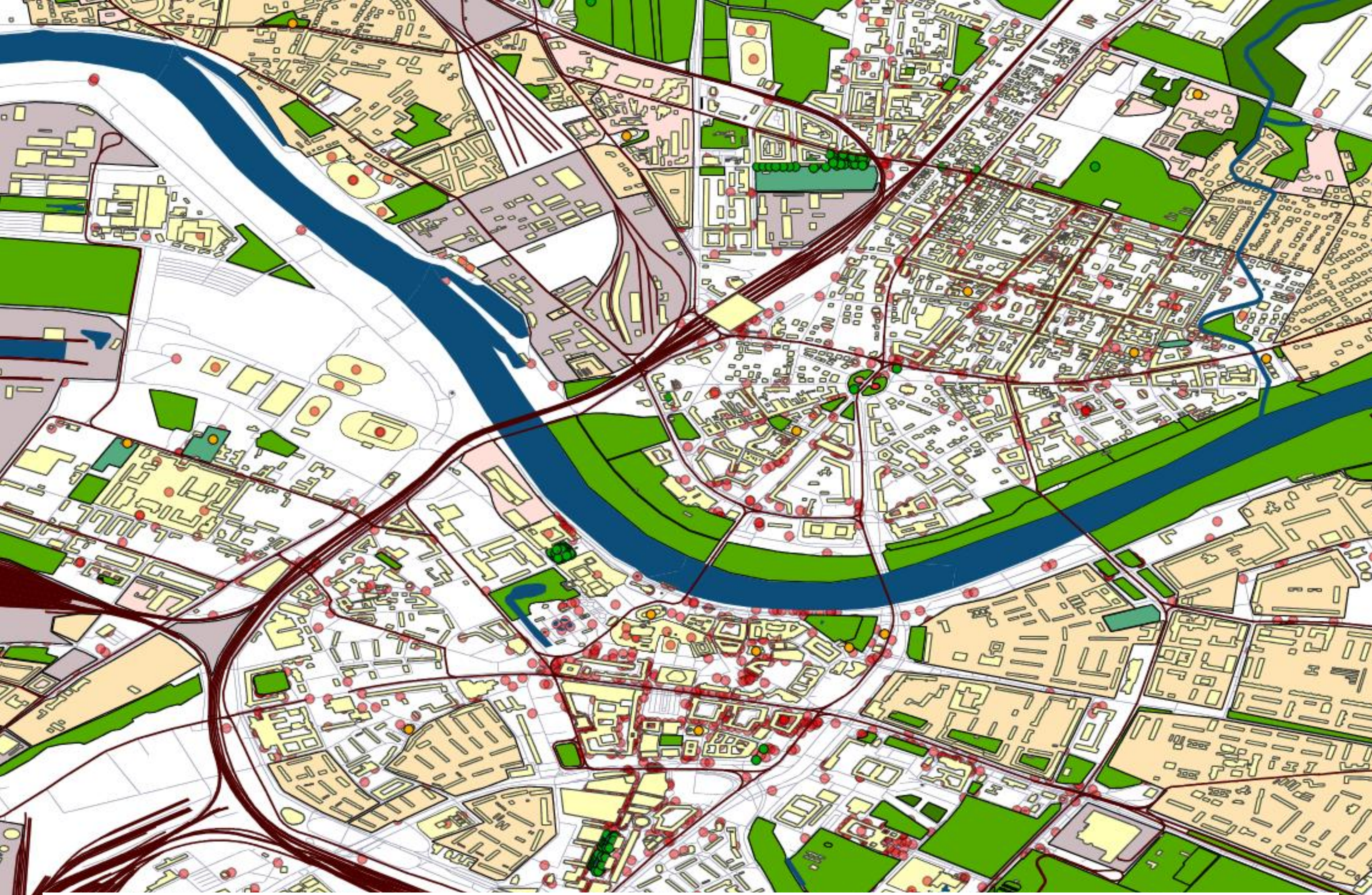
Spaghetti data model – Problems?

- Sharing of common boundary or nodes are not possible
- Spatial relations like adjacency, connectivity, containment etc. are difficult to determine

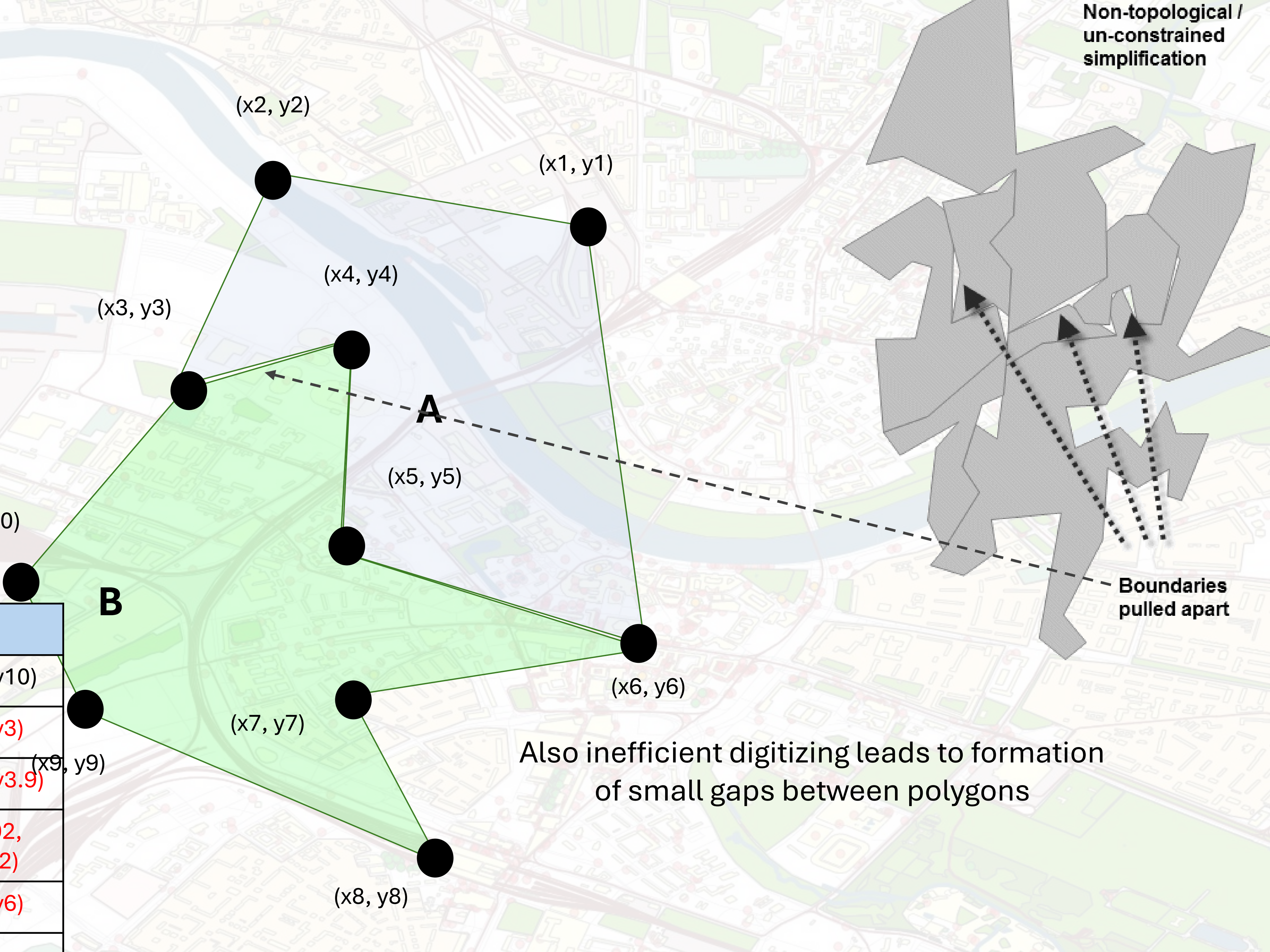


A	B
(x1, y1)	(x10, y10)
(x2, y2)	(x3, y3)
(x3, y3)	(x4, y4)
(x4, y4)	(x5, y5)
(x5, y5)	(x6, y6)
(x6, y6)	(x7, y7)
(x1, y1)	(x8, y8)
	(x9, y9)
	(x10, y10)

shared boundary stored twice



Intuitively, the common boundary problem may not look prominent for two three polygons but for large GIS database with more than 1000’s of polygon, it is a significant issue


Non-topological /
un-constrained
simplification
(x2, y2)
(x1, y1)
(x4, y4)
(x3, y3)
A
(x5, y5)
B
(x6, y6)
(x7, y7)
(x9, y9)
(x8, y8)
Boundaries
pulled apart
Also inefficient digitizing leads to formation
of small gaps between polygons

- Spatial relations like adjacency, connectivity, containment etc. are difficult to determine

Example Case

Lines

a

x	y
1	2
2	3
3	5
3	4
4	4
2	3
2	6
4	7
2	10
1	13

b

x	y
1	7
2	8
4	8
6	7
7	6
9	7
10	5
9	4
8	5
9	1
11	1

c

x	y
10	8
11	5
13	3

Attribute Table for Lines:

Line	*(Attribute – 1)* Road Name	*(Attribute – 2)* Length
a	Road 1	
b	Road 2	
c	Road 3	

Polygons

A

x	y
1	6
2	7
4	7
4	5
5	4
5	1
2	1
1	6

B

x	y
7	4
6	2
5	2
5	3
7	4

C

x	y
9	8
11	9
13	8
12	7
13	4
14	2
10	3
9	5
9	8

- **Topological Data Model**

  - **Topology**, in GIS, is a set of rules to express explicitly the spatial relationship between features

  - The three basic topological relationship are:

    Connectivity: *Arcs are connected at nodes*

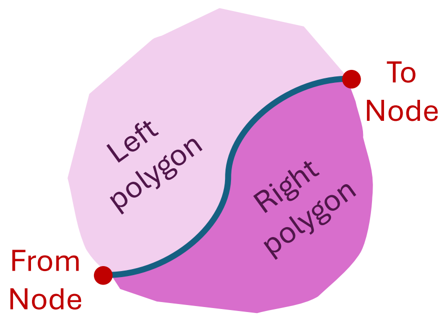
    Area Definition: *Area is defined by a series of arcs*

    Contiguity: *Arcs have directions; and left and right polygon*

  - Spatial relations are easier to determine when compared to spaghetti model

Arc	From Node	To Node
a	B	C
b	A	C
c	C	D
d	C	H
e	L	F
f	D	E
g	D	E
h	D	E
i	F	J
j	F	I
k	F	I
m	E	G
n	I	J
o	J	K

Arcs sharing same node are connected



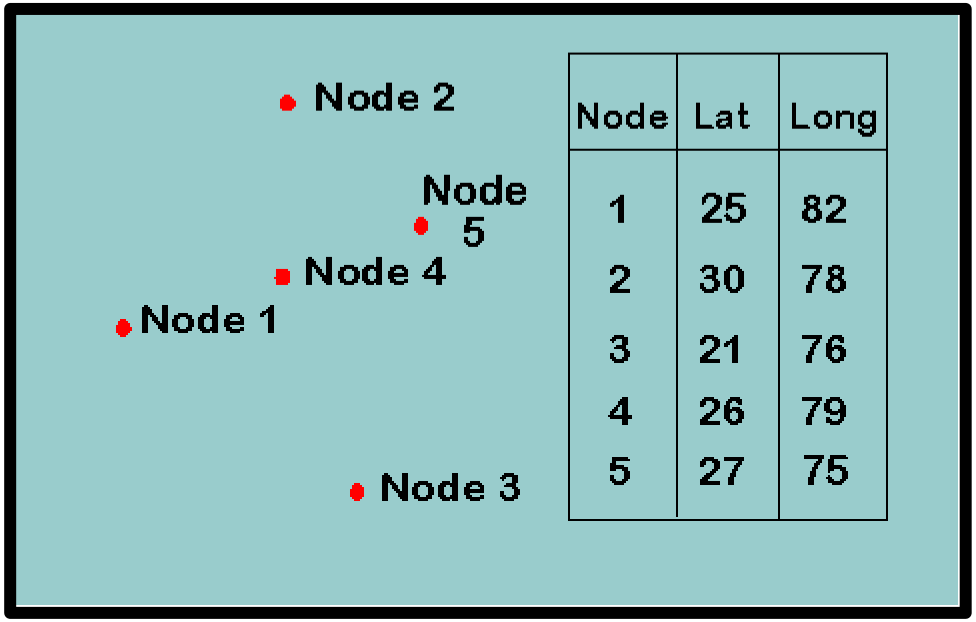
Arcs sharing same node are connected

Example Case

Arc	Left polygon	Right polygon
a	U	U
b	U	U
c	U	U
d	U	U
e	U	U
f	U	I
g	I	II
h	II	U
i	U	IV
j	III	III
k	IV	U
m	U	U
n	U	U
o	U	U

Structure of Nodes in a Topological Model

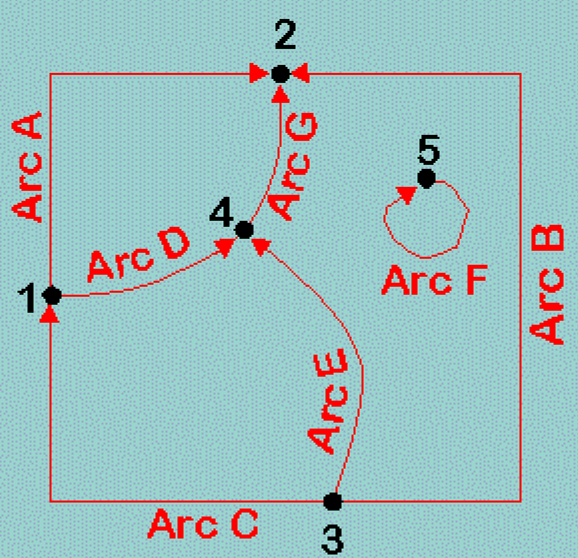
- Points and Nodes are first identified and defined by a pair of x,y coordinates



Node	Lat	Long
1	25	82
2	30	78
3	21	76
4	26	79
5	27	75

Arc-Node Topology

- Nodes are the intersection points where two or more arcs meet
- Apart from the vertices which define an Arc, they will also have a from-node and a to-node list
- Connectivity is defined as all Arcs sharing a common node are connected



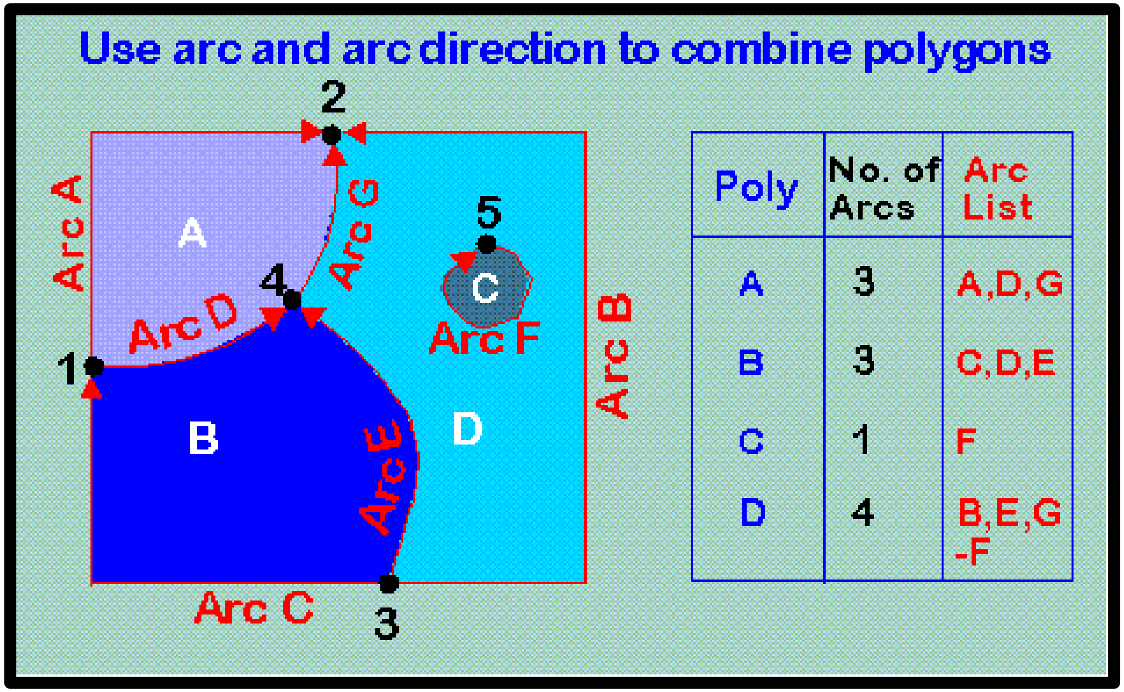
ARC	From Node	To Node
A	1	2
B	3	2
C	3	1
D	1	4
E	3	4
F	5	5
G	4	2

ARC	(x, y) Coordinates
A	(x,y) (x,y) (x,y)....
B	(x,y) (x,y) (x,y)....
C	(x,y) (x,y) (x,y)....
D	(x,y) (x,y) (x,y)....

Polygon-Arc Topology

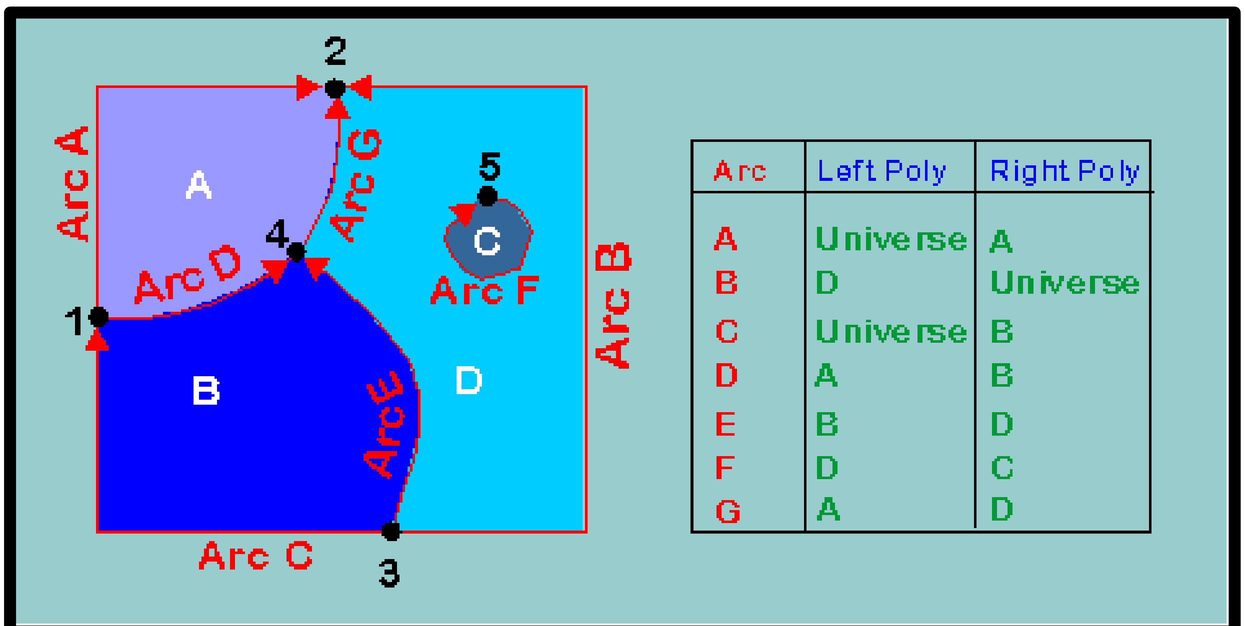
- Topologically connected arcs are used to define the boundary of the polygon feature

- Area is defined by Polygon-Arc Topology



Left-Right Topology

- For each arc, the identity of the polygons to the left and right of it are also recorded
- Contiguity is defined as the polygons that share a boundary are deemed adjacent
- ‘Universe’ is an external area located outside of the study area



Arc	Left polygon	Right polygon
a	U	U
b	U	U
c	U	U
d	U	U
e	U	U
f	U	I
g	I	II
h	II	U
i	U	IV
j	III	III
k	IV	U
m	U	U
n	U	U
o	U	U